

Demonstrating BIT: Battery-free, IC-less and Wireless Smart Textile Interface and Sensing System

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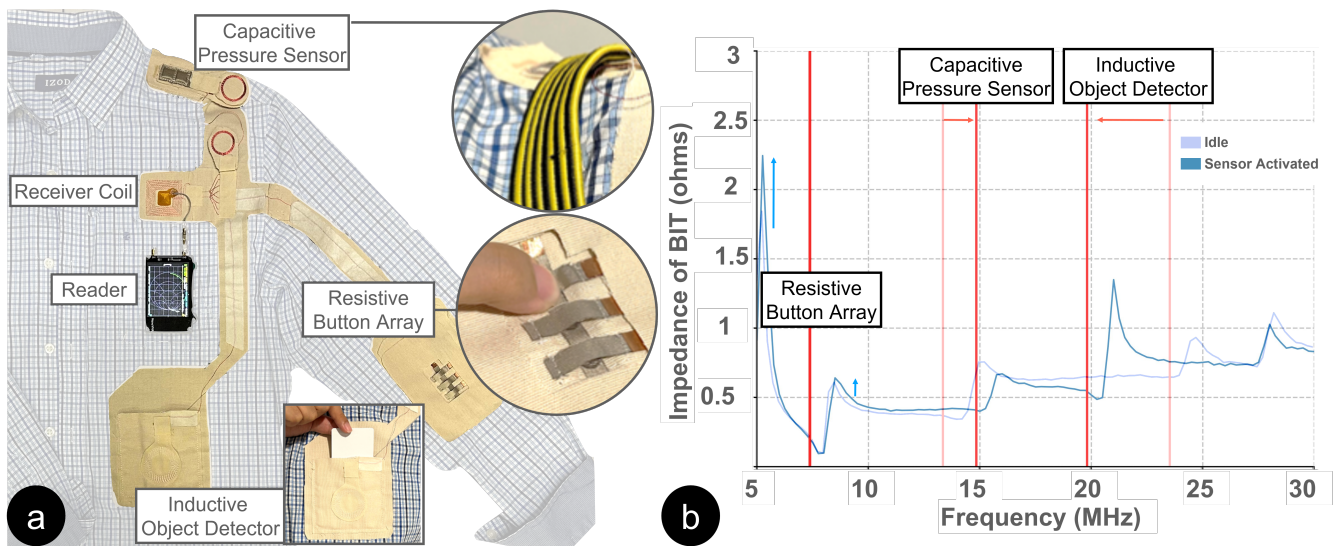


Figure 1: BIT is a smart textile interface that operates without the need for embedding batteries, ICs and connectors into textiles. It features with a receiver coil, enabling wireless coupling and sensor data acquisition with a reader. BIT can be designed to accommodate sensors based on capacitance, resistance, or inductance and support concurrent operations of up to three sensors. (a) For example, a smart shirt was created with BIT, incorporating a resistive button array, a capacitive pressure sensor and an inductive object detector. (b) When interactions occur with the embedded sensors, the impedance spectrum of the smart textile interface changes. This change is wirelessly measured by the reader and can be further interpreted as sensor signals.

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CHI EA '25, Yokohama, Japan

Abstract

The development of smart textile interfaces is hindered by the inclusion of rigid hardware components and batteries within the fabric, which pose challenges in terms of manufacturability, usability, and

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ACM ISBN 979-8-4007-1395-8/25/04
<https://doi.org/10.1145/3706599.3721272>

environmental concerns related to electronic waste. To mitigate these issues, we present a demonstration of BIT: a smart textile interface and its wireless sensing system to eliminate the need for ICs, batteries, and connectors embedded into textiles. BIT is established on the integration of multi-resonant circuits in smart textile interfaces, and utilizing near-field electromagnetic coupling between two coils to facilitate wireless power transfer and data acquisition from smart textile interface. We developed a mathematical model that accurately represents the equivalent circuit of the sensing system, and a novel algorithm to accurately estimate sensor signals based on changes in system impedance. We demonstrate that our technique effectively supports multiple textile sensors of various types.

CCS Concepts

• **Human-centered computing** → **Interaction devices**.

Keywords

battery-less, IC-less, wireless, smart textile, sensing interface, relay

ACM Reference Format:

Weiyu Xu, Tony Li, Yuntao Wang, Xing-Dong Yang, and Te-Yen Wu. 2025. Demonstrating BIT: Battery-free, IC-less and Wireless Smart Textile Interface and Sensing System. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3706599.3721272>

1 Introduction

The development of smart textile technology has enabled integration of daily user input into wearable items such as garments, gloves, and bags, offering an alternative to traditional devices like touchscreens. However, implementing smart textiles faces challenges in manufacturability [8, 22, 33], usability [10, 26, 30], and environmental sustainability [23, 27, 28], primarily due to the embedment of rigid electronic components such as batteries and circuits into textile interfaces. These components compromise comfort, flexibility as well as usability, and contribute to electronic waste when textile products become obsolete.

To address these issues, various methods have been proposed [13, 14, 16, 17, 21, 29, 31, 32], among which resonant sensors hold significant promise [9, 12, 15, 20, 24, 25]. This approach utilizes minimal passive components, typically an LC (inductor-capacitor) resonant circuit consisting of a receiver coil and a capacitive sensor. When the capacitive sensor is pressed, the resonant frequency changes, and this change can be detected through near-field magnetic coupling with a reader. The reader, similar in form factor to a smartphone, can be placed in a pocket, enabling seamless interactions and minimizing the need for embedded electronics in textiles.

However, several challenges remain. First, the approach currently focuses solely on capacitive sensing, limiting its applicability to other wearable sensors that rely on inductance and resistance. Additionally, the design does not account for transmission line deformation or body contact, which can introduce fluctuations in capacitance and inductance, affecting sensor readings. Furthermore, misalignment between the reader and interface coils, common in

wearable contexts, may reduce accuracy. Lastly, the approach supports only single-sensor operation per reader, limiting its usability, as most users carry only one personal device, such as a smartphone.

To overcome these challenges, we proposed BIT, a novel smart textile interface and sensing system. BIT extends the resonator-based technique by involving N-parallel series RLC (resistor-inductor-capacitor) circuits on the receiver coil (Figure 1a). Unlike existing approaches, our method supports up to three different types of sensors commonly used in smart garment applications, including resistive, capacitive, and inductive sensors. Additionally, our approach takes into account the influence of the transmission lines and coil misalignment, ensuring more accurate and robust sensor readings. Furthermore, our technique allows for the concurrent operation of up to three sensors of the same or different types. When a user interacts with the interface, such as pressing a capacitive sensor, the interaction causes a change in the system's impedance (Figure 1b), which can then be wirelessly measured and detected by an external reader. In this paper, we present the design of the BIT system and the sensor value estimation algorithm. Also, based on four types of sensors, we demonstrated two application scenarios: an augmented shirt to detect three kinds of interactions, and an augmented glove to track the bending angle of three fingers.

2 DESIGN OF BIT: BATTERY-FREE, IC-LESS AND WIRELESS SMART TEXTILE INTERFACE

Unlike the traditional LC resonant circuit (Figure 2a), we replaced the capacitor (C) in the typical LC circuit by a series RLC resonant circuit (Figure 2 b), referred as a sensor circuit. Different sensors can be supported by replacing the corresponding components in the sensor circuit. For example, replacing the inductor with a coil forms an inductive sensor for detecting metallic objects. To enable concurrent operation of multiple sensors, we connected multiple sensor circuits with a different operating frequency range in parallel. Based on our pilot study, we designed our model as shown in Figure 2b, mainly following prior research on relay resonators [13].

During operation, user interaction with the sensor induces a change in the value of the corresponding component within the RLC resonant circuit (e.g., inductance variations due to metallic objects), ultimately resulting in changes in the impedance spectrum within the reader circuit through magnetic coupling. Our system measured these changes and used an algorithm to extract sensor values for the detection of user input.

3 Sensor Value Estimation

Two challenges hinder the extraction of the sensor value from the reader: (1) the model involved too many unknown parameters, such as k , C_{line} and each sensor value, and (2) mobile impedance readers were typically limited in capturing a high-resolution, high-precision impedance spectrum. Thus, we propose a three-step algorithm to estimate the sensor value from the impedance spectrum captured by the reader.

3.1 Step 1: Calculating the coupling factor

At resonance, the impedance of the sensor circuit is near zero if the resistance is low, which makes the coupling factor (k) the only

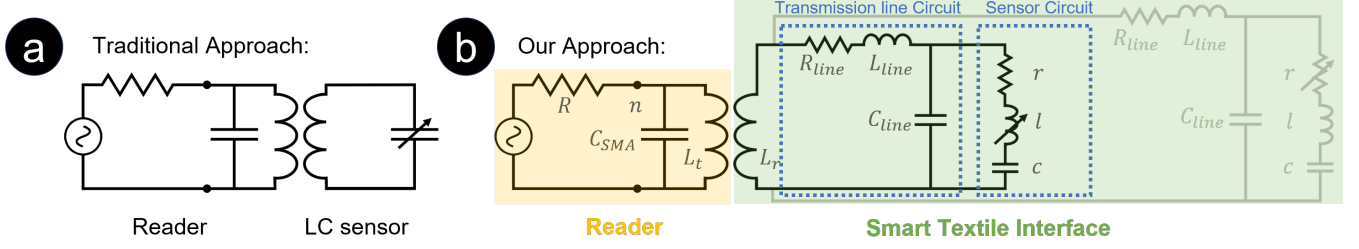


Figure 2: (a) The equivalent circuit of a traditional resonant sensor system. (b) The equivalent circuit of our entire system, which consists of a reader circuit and a smart textile interface circuit. The interface includes a receiver coil (L_r) connected in parallel with multiple transmission and sensor circuits. Each transmission line circuit contains parasitic resistance (R_{line}), inductance (L_{line}), and capacitance (C_{line}), connected to a sensor circuit with a resistor (r), capacitor (c), and inductor (l) in series. The reader circuit includes a voltage exciter, internal load (R), and transmitter coil (L_t), with parasitic capacitance (C_{SMA}). Impedance is measured by capturing voltage at point (n) and applying the voltage divider rule.

unknown parameter in calculating the magnitude of impedance. In light of this, we additionally incorporated a reference circuit with known LC components into the smart textile interface to directly calculate k value. Due to the limited number of data points measured from the reader, a linear interpolation technique was used to estimate the impedance at the desired frequency.

3.2 Step 2: Estimating Capacitive or Inductive Sensor Values

As k is known, the theoretical absolute impedance spectrum when each sensor circuits is shorted can be plotted, and the resonant frequency can be retrieved from the intersection between the measured impedance spectrum and the theoretical impedance spectrum, as shown in figure 3. The algorithm seeks for the points where the impedance trend increases as it indicates that the system is approaching to another resonance. Once the resonant frequency is determined, the algorithm will derive the capacitive or inductive sensor value using the formula of the resonant frequency [19]

3.3 Step 3: Approximating Transmission Line Capacitance and Resistive Sensor Values

Unlike the previous steps, no alternative methods were available to estimate the resistive sensor values. The algorithm firstly estimated C_{line_i} by adjusting each C_{line_i} until the lowest peaks' frequencies from the predicted S11 parameter spectrum align with those from measure S11 spectrum. Then, the algorithm shifted to approximate resistive sensor values using a regression fit optimization algorithm.

4 IMPLEMENTATION

4.1 Battery-free, IC-less and Wireless Smart Textile Interface

The smart textile interface of BIT, consisting of a receiver coil, transmission lines, constant RLC components, and textile sensors, was implemented on the Muslin Fabric Cotton substrate [3]. 34AWG Litz Wire [1], a common choice for low-resistance and insulated wiring [7, 11, 18] were opted to fabricate the receiver coil and transmission lines. The wire was embroidered as bobbin thread through straight

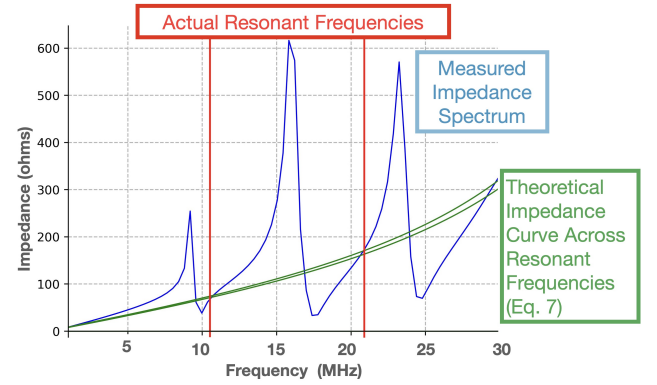


Figure 3: The absolute impedance spectrum of a prototype with three resonant sensor circuits (10.6 MHz, 20.9 MHz, and 30.9 MHz) and the supposed resonant frequencies (red line) of each sensor circuit of a prototype. The green lines illustrate the value of theoretical absolute impedance spectrum using the estimated coupling factor (k) calculated by the two sensor circuits with lower resonant frequency. Their intersections with the measured impedance spectrum accurately represent the resonant frequencies of the sensor circuits.

stitches with the stitch length of 2.5mm through Brother SE600 embroidery machine [6]. For constant RLC components, 2-pin 0604 SMD components were soldered onto the Litz wires, following the method described in [22]. Embroidered or I-shaped coil inductors were also used for higher quality factor. For sensors, four representatives were implemented and demonstrated in Figure 4.

4.2 Reader and Operating System

The reader of BIT system consisted of a NanoVNA, connected to a transmitter coil measuring 10mm by 10mm, as shown in Figure 1. The reader is designed to perform sweeps of frequencies ranging from 5 MHz to 30 MHz with a total of 101 sampling points and a

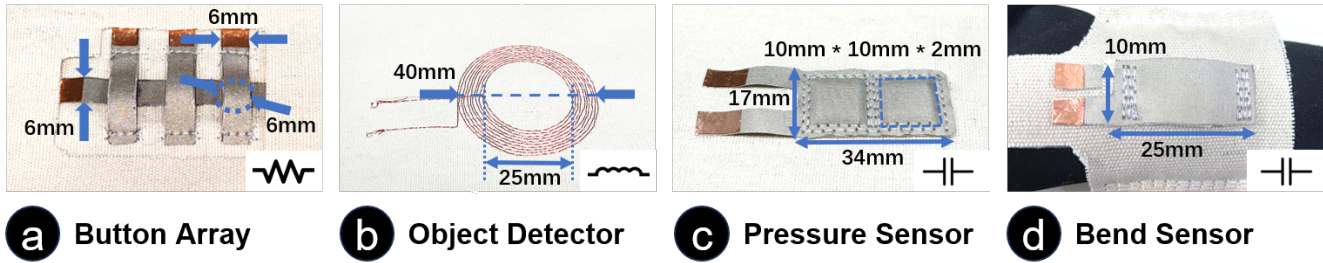


Figure 4: Four representative textile sensors implemented in our battery-free and IC-less smart textile interface. (a) A resistive button array consisting of 3 pressure-sensitive buttons. Each button is constructed from two pieces of conductive fabric strip (width = 6mm)[4] sandwiching a circular piece of resistive foam (thickness = 1mm)[5]. The resistance of the button array is 60Ω with no pressure. When each button is pressed, the resistance will drop to 0Ω to 15Ω , 15Ω to 30Ω and 30Ω to 45Ω respectively according to the pressure applied. (b) An inductive object detector consisting of a 10-turns coil (25mm by 40mm) with an inductance of 5.4 uH. The inductance will decrease accordingly when metallic object (e.g. a key, a card embedded with coils) approaches. (c) A capacitive pressure sensor constructed by two pieces of conductive fabric (17mm by 34mm) sandwiching two $10\text{mm} \times 10\text{mm} \times 2\text{mm}$ square pieces of resistive rubber [2]. The two conductive layers are isolated by a layer of cotton fabric [3]. The default capacitance is 15pF and will increase to 20 pF when 2kg pressure is applied. (d) a capacitive bend sensor constructed by two pieces of conductive fabric (10mm by 25mm) sandwiching a piece of resistive foam (thickness = 1mm). The two conductive layers are isolated by a layer of cotton fabric. When the sensor is bent from 0 to 90 degrees, the capacitance will increase from 7pF to 15pF.

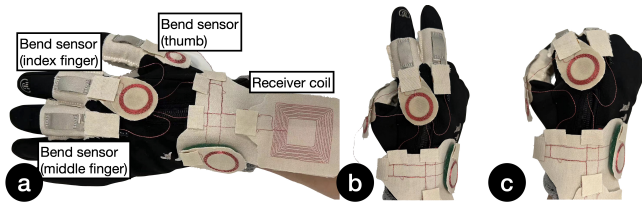


Figure 5: Our glove prototype integrated with BIT. (a) The glove features three bend sensors on the thumb, index, and middle fingers to capture finger gestures. In our implementation, for example, the glove can detect (b) peace and (c) fist gestures.

sampling rate of 10 Hz. Data were passed to a laptop (Thinkpad Carbon X1) to process the signals.

4.3 Example Applications

4.3.1 Smart Shirt. We incorporated an inductive object detector, a capacitive pressure sensor, and a resistive button array into a battery-less and IC-less smart textile interface on a shirt, as shown in figure 1a. This shirt has several applications, including health tracking, where a smartphone app monitors shoulder pressure and reminds the user to relieve it, preventing strain or pain. It can also detect metallic objects, like keys or access cards, in the pocket and trigger notifications if they're left behind. Additionally, the shirt features shortcut buttons on the sleeve for controlling music playback or interacting with an AI assistant.

4.3.2 Smart Glove. We integrated the smart textile interface into a glove, placing the receiver coil in a position that corresponds to where a smartwatch would typically be situated on the wrist (Figure

5a). We incorporated three capacitive bending sensors on the glove to capture the gestures of thumb, index and middle fingers. This glove can serve as an extension of the input device on the user's smartwatch, allowing users to control functions like answering calls with simple gestures such as peace and fist (Figure 5b and 5c). Furthermore, the integration of the glove can enhance VR and AR experiences by empowering users to engage in immersive interactions without having to worry about charging the glove.

Acknowledgments

This work is supported by the Tsinghua University Initiative Scientific Research Program, the Undergraduate Education Innovation Grants and Florida State University First Year Assistant Professor (FYAP) Grant.

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